Public Level 2 EV Charging Location Planning in New York City

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1. Problem Definition

Our project focuses on a single investment cycle for additional public Level 2 (L2) charging ports at existing NYC DOT municipal parking lots and garages. We consider two types of investment: adding ports at facilities that already offer public L2 charging and installing the first public L2 ports at facilities without them. Building new parking facilities, acquiring land, and investing in curbside chargers, privately owned facilities, or DC fast chargers are outside the investment scope.

The intended decision maker is the infrastructure planning team within the New York City Department of Transportation (NYC DOT). Our project evaluates a hypothetical investment scenario rather than an officially allocated municipal charging budget.

We will formulate a capacitated location-allocation mixed-integer linear program to maximize an equity-weighted measure of estimated weekly public L2 charging energy served, measured in kWh/week. The model will use a hypothetical capital budget of $3.75 million, subject to site installation limits, charging capacity, geographic accessibility, and projected demand. In the initial model, demand will be represented by one geographic point per reviewed ZIP area and may be served by charging sites within an assumed 0.5-mile radius. Equity weights will be computed before optimization from the existing EV-registration and public operational L2 charging datasets.

2. Assumptions List

1. We assume that $3.75 million is available for additional municipal L2 charging projects, independently of existing committed projects. This amount is a planning assumption of the allocation of the $15 million CFI grant NYC DOT received [1]. To inform the scale of this scenario, we use North Carolina's historical allocation of 25% of its zero-emission-vehicle infrastructure funding to L2 charging as a policy benchmark [2].

2. Demand locations are estimated before optimization because exact EV addresses are unavailable. The base service radius is 0.5 mile using geodesic distance with documented barrier exclusions. A ZIP representative point is an approximation; finer spatial placement and alternative radii provide sensitivity cases.

3. Capacity and feasibility: Each port has a fixed effective weekly capacity after accounting for operating hours, downtime, and shared power. One independently usable Level 2 port serves one charging space. Each site has a maximum additional-port limit reflecting space and assumed electrical feasibility. Unknown inventory counts are not treated as zero.

4. Estimated weekly public Level 2 demand uses EV registrations by ZIP code, estimated energy use per vehicle, and an assumed public Level 2 charging share. Demand may vary across different vehicles. Projected demand is represented by $D_i(1+r)^T$, where r and T are scenario assumptions rather than values estimated from the available cross-sectional data.

5. Charging-accessibility equity is represented by a baseline supply-shortage weight g_i, computed from the existing EV-registration and public operational L2 charging datasets. The policy weight λ is a chosen, dimensionless scenario parameter and is not data-derived. Baseline equity weights are held fixed when demand-growth scenarios are compared.

6. We use the ICCT 2019 charging hardware and installation cost as benchmarks rather than 2026 NYC costs. We assume cost can be split into fixed site-level costs (F_j) and incremental costs (c_j) per additional port.

3. Sets and indices

Set	Index	Definition
Z	$z \in Z$	Reviewed ZIP-based demand areas; document the mailing-ZIP to ZCTA mapping.
I	$i \in I$	Demand points fixed before optimization; use one point per ZIP for an initial case or smaller weighted areas in a finer case.
M	$j \in M$	Eligible existing DOT municipal parking facilities where public Level 2 ports can be added.
M_E and M_N		M_E = Facilities in M that have existing public Level 2 ports. M_N = Facilities in M that have no existing public Level 2 ports.
O	$j \in O$	Other existing publicly accessible Level 2 sites, including curbside and public-access private stations. No investment is allowed here.
$J = M \cup O$	$j \in J$	Set of all modeled charging supply sites; each physical charging resource appears only once.
A	$(i, j) \in A$	Set of accessible pairs (i,j) with $d_{ij} \leq R$, after documented barrier exclusions.

4. Optimization Parameters

Symbol	Definition and units	Base case or estimation basis
B	Available capital, dollars	$3,750,000; sensitivity $2.25M and $5.25M. Based on DNV's 2022 average municipal Level 2 project cost of $11,687 per port, an illustrated expansion of 320 ports would cost approximately $3.74 million.
D_i	Baseline public Level 2 energy demand at point *i*, kWh/week	Estimated from 2026 NYS DMV registrations by ZIP code, weekly energy use per EV, and assumed public L2 charging share.
r; T	Annual demand growth rate; planning horizon in years	Assumed scenario inputs selected before optimization. No numerical values are claimed as data-derived in this milestone.
d_{ij}; R	Distance from demand point *i* to supply point *j*, miles	Calculated from ZCTA (converted in set Z) coordinates; R = 0.5 mile (the maximum distance).
n^0_j	Existing public Level 2 ports at site *j*, ports	AFDC inventory: 3508 reported ports across 758 public operational L2 station records.
C^0_j	Effective existing Level 2 capacity, kWh/week	Estimated using charging power, operating hours, and shared-power limits; *j* in J.
S^0_i	Accessible existing L2 supply near point *i*, kWh/week	Computed from the public operational L2 dataset as the sum of C^0_j over accessible sites. It is an accessibility proxy and may include sites shared by multiple demand points.
g_i	Baseline charging-supply shortage score, unitless	For $D_i > 0$, $\max(0, 1 - S^0_i/D_i)$; otherwise 0. Computed before optimization from existing datasets and bounded from 0 to 1.
λ	Equity-policy weight, unitless	Chosen scenario assumption; test $\lambda = 0$ as the unweighted benchmark and report sensitivity for any positive value used.
q_j	Capacity per added port, kWh/(port*week)	Estimated using charging power, operating hours, and shared-power limits; *j* in M.
U_j	Maximum additional ports	Estimated from parking-space data for 39 municipal facilities and assumed electrical limits; *j* in M.
F_j	Fixed installation cost for site without previous L2 ports, dollars/site	Baseline assumption: F_j = $20,000 for first-time installations requiring shared electrical upgrades [5].
a_j	First-time installation indicator, unitless	Known based on verified inventory; 1 if facility *j* does not have L2 ports; 0 if it already has L2 ports.
c_j	Incremental installed cost, dollars/port	Based on ICCT 2019 hardware and installation benchmarks, adjusted for cost assumptions without double-counting F_j; *j* in M [6].

For the Demand Estimation, we estimated baseline weekly public Level 2 charging demand using NYS DMV registrations by ZIP code. For example, ZIP 10001 has 160 normal vehicles, 2 taxis, and other types of records. For Capacity Estimation, we estimate C_i^0 from existing charging stations, effective charging power, weekly operating hours, and availability, using the same approach for expanded ports. The baseline equity score g_i compares accessible existing public L2 capacity with baseline demand and does not require an additional dataset. Future demand uses the multiplier $(1+r)^T$, where r and T are declared scenario assumptions. For cost estimation, we use ICCT 2019 benchmarks, including $3,127 for networked single-charger hardware and $3,090 for installation.

5. Decision Variables

There are three decision variables that determine investment locations, the number of additional Level 2 ports, and weekly charging demand allocation.

Variable	Type	Definition	Unit	Domain
y_j	Binary variable	Equals 1 if additional Level 2 ports are installed at municipal facility (j), and 0 otherwise.	Binary (unitless)	$y_j \in \{0,1\}$, $\forall j \in M$
x_j	Nonnegative integer variable	Number of additional public Level 2 charging ports installed at municipal facilities (j).	Ports	$x_j \in \mathbb{Z}_{\geq 0}$, $\forall j \in M$
f_{ij}	Nonnegative continuous variable	Weekly public Level 2 charging demand at point (i) served by charging site (j).	kWh/week	$f_{ij} \in \mathbb{R}_{\geq 0}$, $\forall (i,j) \in A$

6. Constraints

6.1 Capital budget

$$\sum_{j \in M} (F_j a_j y_j + c_j x_j) \leq B \tag{1}$$

The total fixed site costs and per-port costs must not exceed the budget (B). This investment budget does not include the historical installation costs for existing ports. Source: B = $3.75 million in the base case ($2.25 million / $5.25 million for sensitivity analysis), an assumption. Each added port incurs cost c_j. The additional one-time setup cost F_j is incurred only when a facility both has no existing L2 ports and receives investment.

6.2 Investment activation and installation limit

$$y_j \leq x_j \leq U_j y_j, \qquad \forall j \in M \tag{2}$$

Only selected municipal facilities can add ports. When y_j=0, x_j=0. When y_j=1, the newly added quantity must be between 1 and U_j. U_j is estimated from spaces available for new charging (NYC DOT Municipal Parking Facilities data, net of existing charging spaces) and an assumed electrical limit.

6.3 Projected demand limit

$$\sum_{\substack{j \in J \\ (i,j) \in A}} f_{ij} \leq D_i (1+r)^T, \qquad \forall i \in I \tag{3}$$

The total amount of charging service allocated to demand point i cannot exceed its projected demand $D_i(1+r)^T$. Demands at the same point can be jointly met by multiple accessible sites. Unmet demand is allowed, and it is not required that all demand be met. The baseline D_i is estimated from NYS DMV EV registrations by ZIP code, times assumed weekly energy per EV and the share of charging done at public Level 2. The parameters r and T are selected before optimization and are explicitly treated as scenario assumptions, so the right-hand side remains constant and the model remains linear.

6.4 Capacity at municipal facilities

$$\sum_{\substack{i \in I \\ (i,j) \in A}} f_{ij} \leq C_j^0 + q_j x_j, \qquad \forall j \in M \tag{4}$$

Energy served cannot exceed existing capacity plus q_j per added port. Source: n^0_j is obtained from AFDC and municipal records; $C^0_j = 0$ only for facilities verified to have no publicly accessible Level 2 ports (M_N). C^0_j and q_j are estimated from charger power, public hours, and assumed downtime and shared-power limits. Observed usage (NYC Municipal EV Charging Data) is a plausibility check, not a capacity cap.

6.5 Capacity at other existing sites

$$\sum_{\substack{i \in I \\ (i,j) \in A}} f_{ij} \leq C_j^0, \qquad \forall j \in O \tag{5}$$

Curbside and other existing public Level 2 sites can only serve requests within the existing capacity. This project does not allow the addition of new ports on these sites. Source: Locations, port counts, and hours from the AFDC Station Locator, converted to kWh/week with the same capacity-estimation method as in Section 6.4, using site-specific inputs where available.

6.6 Accessibility and domains

$$\begin{aligned} &f_{ij} \in \mathbb{R}_{\geq 0}, && \forall (i,j) \in A, \\ &x_j \in \mathbb{Z}_{\geq 0}, \quad y_j \in \{0,1\}, && \forall j \in M. \end{aligned} \tag{6}$$

Flow variables f_{ij} are defined only for pairs in the accessible-pairs set A. Pairs included in A must satisfy $d_{ij} \leq 0.5$ mile and pass the barrier-exclusions screening. Therefore, demand can only be allocated to sites within the distance threshold that meet the accessibility conditions. Source: from geocoded facility/station addresses and ZIP representative points.

7. Objective Function

$$\max \sum_{(i,j) \in A} (1 + \lambda g_i) f_{ij}$$

The revised objective maximizes an equity-weighted measure of estimated weekly public Level 2 EV charging demand served by the charging network. The original total-demand term is preserved, while the additional term λ multiplied by the sum of $g_i f_{ij}$ over all (i,j) in A gives greater priority to demand areas with relatively insufficient accessible existing public L2 supply.

To compute the equity input from the existing datasets before optimization, define:

$$S_i^0 = \sum_{\substack{j \in J \\ (i,j) \in A}} C_j^0, \qquad g_i = \max\left(0, 1 - \frac{S_i^0}{D_i}\right) \quad \text{for } D_i > 0$$

Set $g_i = 0$ when $D_i = 0$. Here, D_i comes from the EV-registration demand estimate and S^0_i comes from accessible public operational L2 capacity. Because a charging site may be accessible to multiple ZIP demand points, S^0_i and g_i are supply-accessibility proxies, not exact measures of exclusive capacity or unmet demand. Network capacity constraints (4) and (5) continue to prevent physical capacity from being allocated more than once.

The parameter λ controls the strength of the equity priority. λ=0 reproduces the original unweighted objective. Any positive λ used in the implementation must be reported as a policy scenario assumption and evaluated with sensitivity analysis; it is not inferred from the datasets. The baseline g_i is held fixed across values of r and T so that projected demand growth does not get mixed into the definition of existing charging-resource shortage.

The symbols are: i, demand point, i in I; j, charging supply site, j in J; A, set of accessible demand-site pairs satisfying the 0.5-mile distance threshold and other accessibility conditions; f_{ij}, weekly charging demand at point i served by site j (kWh/week); g_i, baseline charging-supply shortage score; and λ, the chosen equity-policy weight.

By summing the equity-weighted accessible f_{ij}, the objective evaluates the EV demand the whole charging network can meet, including existing and expanded charging infrastructure, while favoring areas with greater baseline charging-supply shortage. This objective is appropriate for NYC DOT since our focus is not just to install as many EV charging ports as possible, but to use a limited investment budget to achieve an effective and more equitable allocation of charging resources while satisfying the budget, installation feasibility, accessibility, projected demand, and charging-capacity constraints. The weighting provides a priority mechanism; it does not guarantee equal service rates across all ZIP areas.

References

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